Name: _____

TOPIC TEST 1

Chapter 10: Statistical inference

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 8 multiple-choice questions (8 marks).
- Part B: 9 short-answer questions (30 marks).
- Part C: 2 analysis questions (22 marks).

Part A

Multiple-choice questions

8 multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 What is the mean and standard deviation of the results below?

12, 9, 13, 6, 11, 8, 12, 9, 11, 10, 12, 10

A 10.25, 2.03

B 11.12, 1.92

C 10.25, 3.69

D 11.12, 4.04

E 10.25, 1.92

[1 mark]

2 A sample of 36 items has a mean of 28 and a standard deviation of 9. What is the standard deviation of the sampling mean?

A 0.88

B 1.5

C 1.76

D 1.70

E 3.11

[1 mark]

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3 The sampling distribution of the mean mass of hands of Cavendish bananas was found to have a mean of 2.3 kg with a standard deviation of 0.1 kg for samples of 40 hands. What would be the standard deviation of individual hands?

A 0.1 kg

B 4 kg

C 0.63 kg

D 2 kg

E 0.15 kg

[1 mark]

4 The average order at a takeaway shop is $22.59 with a standard deviation of $8.21. What is the probability that the next 10 orders will total more than $250?

A 0.2287

B 0.1766

C 0.3846

D 0.8234

E 0.3234

[1 mark]

5 A call centre servicing a national pizza chain has an average call length of 88 seconds, with a standard deviation of 20 seconds. What is the probability that an operator's next 6 calls have an average length of between 90 and 100 seconds?

A 0.5968

B 0.9292

C 0.4292

D 0.1711

E 0.3324

[1 mark]

6 What is the 95% confidence interval for a normal variable with mean 425 and standard deviation 62?

A [303, 547]

B [265, 585]

C [366, 484]

D [323, 527]

E [268, 582]

[1 mark]

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7 What is the confidence level for the interval 123–137 for a normal variable with mean 130 and standard deviation 4?

A 90%

B 96%

C 99%

D 92%

E 95%

[1 mark]

8 What is the 99% confidence interval for the population mean if the mean of a sample of 50 is 34 with a standard deviation of 9?

A [10.9, 57.1]

B [38.7, 61.3]

C [20.9, 79.1]

D [32.7, 35.3]

E [30.7, 37.3]

[1 mark]

Part B

Short-answer questions

9 short-answer questions
30 marks
Show your working where appropriate.

1 **a** A normal coin is tossed 10 times and the number of heads is noted. This experiment is repeated 1000 times. What is a graph of the results likely to look like?

b Instead of graphing the results directly, the average numbers of heads in groups of 40 sets of 10 is calculated and these averages are graphed. What is this graph likely to look like?

[4 marks]

2 **I** A sample of 30 is chosen from a binomial distribution with $p = 0.7$ and $n = 20$ and a graph of the sample is drawn.

II 20 samples of 30 are chosen from a binomial distribution with $p = 0.7$ and $n = 20$ and a graph of the means of each sample is drawn.

III 20 samples of 80 are chosen from a binomial distribution with $p = 0.7$ and $n = 20$ and a graph of the means of each sample is drawn.

How will the graphs from parts **I**, **II** and **III** be likely to compare?

[6 marks]

3 What is the mean and standard deviation of a sampling distribution of a binomial distribution with $p = 0.35$ and $n = 32$ for samples of 45?

[3 marks]

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4 What is the mean and standard deviation of a sampling distribution of a uniform distribution on the interval [15, 33] for samples of 40?

[3 marks]

5 The average mass of a box of 40 peaches is 11.6 kg with a standard deviation of 0.2 kg. Given that the cartons and packing weight 500 g, what would you expect the average mass and standard deviation of individual peaches to be?

[3 marks]

6 What is the 93% confidence interval for the standard normal variable Z?

[2 marks]

7 What is the 90% confidence interval for the normal variable X with mean 42 and variance 15?

[3 marks]

8 What is the confidence level for the interval [40, 60] for a normal variable with a mean of 50 and a variance of 18?

[3 marks]

9 What is the 95% confidence interval for the population mean if a sample of 40 had a mean of 35 and a standard deviation of 6?

[3 marks]

Part C

Analysis questions

2 analysis questions
22 marks
Show your working where appropriate.

1 From a survey of 600 people, it was reported that the 97% confidence interval for the weekly amount spent on transport was \$75.60 to \$84.90. What is the probability that, if you asked another 12 people, they would spend more than a total of \$1000 a week on transport?

[8 marks]

2 The populations of Australian states and territories are:

New South Wales	7.5 million
Victoria	5.8 million
Queensland	4.7 million
Western Australia	2.5 million
South Australia	1.7 million
Tasmania	0.5 million
Australian Capital Territory	0.4 million
Northern Territory	0.3 million

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A national survey of the amounts Australians spent on mobile phones found it was $340 a month with a margin of error of 5%. The survey was taken by phoning a stratified sample of 1200 people on the electoral rolls from the capital cities and asking the 'head of the household'. The calculated variances for each capital city were approximately the same.

a What does the stated margin of error mean?

b How many from each capital were interviewed?

c What are the margins of error for the individual states and territories?

d State any problems with the survey.

[14 marks]

[Total marks: 60]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $\Sigma x = 123$, so mean $= 10.25$, $\Sigma x^2 = 1305$ so $\sigma = \sqrt{\frac{1305}{12} - 10.25^2} = 1.920...$

$\therefore$ **E** **[1 mark]**

2 $\frac{9}{\sqrt{36}} = 1.5$

$\therefore$ **B** **[1 mark]**

3 $0.1 \times \sqrt{40} = 0.632\ldots$

$\therefore$ **C** **[1 mark]**

4 $\text{normCDf}\left(25, \infty, 22.59, \frac{8.21}{\sqrt{10}}\right) = 0.176\,339\ldots$

$\therefore$ **B** **[1 mark]**

5 $\text{normCDf}\left(90, 100, 88, \frac{20}{\sqrt{6}}\right) = 0.332\,425\ldots$

$\therefore$ **E** **[1 mark]**

6 $1.96 \times 62 = 121.52$, $425 - 121.52 = 303.48$, $425 + 121.52 = 546.52$

$\therefore$ **A** **[1 mark]**

7 $\text{normCDf}(123, 137, 130, 4) = 0.919\ldots$

$\therefore$ **D** **[1 mark]**

8 $2.576 \times \frac{9}{\sqrt{50}} = 3.278\ldots$, $34 \pm 3.278\ldots \approx [30.7, 37.3]$

$\therefore$ **E** **[1 mark]**

Part B – Short-answer questions

1 a It is symmetrical around the peak of 5, gradually tapering off towards 0 and 10. The standard deviation is about 1.6. **[2 marks]**

b It is symmetrical around the peak of 5, but tapers off quickly to about 4 and 6. The standard deviation is about 0.25. It is more concentrated than **a**. **[2 marks]**

[4 marks altogether]

2 I The graph is skewed to the left, with a mean of 14 and a standard deviation of about 2. **[2 marks]**

II The graph is less skewed than **I**, with the same mean but a standard deviation of only about 0.32. It is much more concentrated than **I**. **[2 marks]**

III The graph is almost symmetrical, with the same mean, 14, but a standard deviation of only about 0.16 and is very concentrated in the centre. **[2 marks]**

[6 marks altogether]

3 Mean $= 0.35 \times 32 = 11.2$, **[1 mark]**

$\text{SD} = \sqrt{\frac{32 \times 0.35 \times 0.65}{45}} \approx 0.402$ **[2 marks]**

[3 marks altogether]

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4 Mean $= 0.5(15 + 33) = 24$, **[1 mark]**

SD $= \dfrac{33 - 15}{\sqrt{12} \times \sqrt{40}} \approx 0.822$ **[2 marks]**

[3 marks altogether]

5 Average mass of 1 peach $= 11.5 \text{ kg} \div 40 = 287.5 \text{ g}$ **[1 mark]**

Standard deviation of average mass of 40 peaches $= 0.2 \div 40 = 5 \text{ g}$ **[1 mark]**

Standard deviation of individual mass $= 5 \times \sqrt{40} \approx 31.6 \text{ g}$ **[1 mark]**

[3 marks altogether]

6 invNormCDf(0.035,0,1) ≈ -1.812 **[1 mark]**

93% confidence interval is $[-1.812, 1.812]$ **[1 mark]**

[2 marks altogether]

7 $42 - 1.645 \times \sqrt{15} \approx 35.6$ **[1 mark]**

$42 + 1.645 \times \sqrt{15} \approx 48.4$ **[1 mark]**

90% confidence interval is [35.6, 48.4] **[1 mark]**

[3 marks altogether]

8 normCDf(40,60,50, $\sqrt{18}$) $= 0.9815\ldots$ **[2 marks]**

The confidence level is about 98.1% **[1 mark]**

[3 marks altogether]

9 SD $= \dfrac{6}{\sqrt{40}} = 0.948\ldots$ **[1 mark]**

Margin of error $= 1.96 \times 0.948\ldots = 1.858\ldots$ **[1 mark]**

95% confidence interval $=$ [33.14, 36.86] **[1 mark]**

[3 marks altogether]

Part C – Analysis questions

1 For the 97% confidence interval, $z =$ invNormCDf(0.985,0,1) $= 2.170\ldots$ **[1 mark]**

Margin of error $= \dfrac{84.9 - 75.6}{2} = 4.65$ **[1 mark]**

Sampling SD $= 4.65 \div 2.17\ldots = 2.142\ldots$ **[1 mark]**

SD of weekly amount $= 2.142\ldots \times \sqrt{600} = 52.48\ldots$ **[1 mark]**

SD of sample of 12 $= \dfrac{52.48...}{\sqrt{12}} = 15.15\ldots$ **[1 mark]**

Mean $= \dfrac{75.6 + 84.9}{2} = 80.25$ **[1 mark]**

normCDf($\dfrac{1000}{12}$, ∞, 80.25, 15.15…) $= 0.419\ldots$ **[1 mark]**

If you asked another 12 people, the probability that they would spend more a total of more than $1000 a week on transport is about 0.42 **[1 mark]**

[8 marks altogether]

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2 a It means that the 95% confidence interval margin of error is 5% of \$340 = \$17 **[1 mark]**

b Total population = 23.4 million **[1 mark]**

Numbers interviewed:

New South Wales $= \frac{7.5}{23.4} \times 1200 \approx 385$

Victoria $= \frac{5.8}{23.4} \times 1200 \approx 297$

Queensland ≈ 241

Western Australia ≈ 128

South Australia ≈ 87

Tasmania ≈ 26

Australian Capital Territory ≈ 21

Northern Territory ≈ 15 **[4 marks]**

c Standard deviation of amount spent $= 17 \div 1.96 \times \sqrt{1200} = \$300.46$ **[1 mark]**

95% margins of error:

New South Wales $= 1.96 \times \frac{300.46}{\sqrt{385}} \approx \$30.01$

Victoria $= 1.96 \times \frac{300.46}{\sqrt{297}} \approx \$34.17$

Queensland ≈ \$37.93

Western Australia ≈ \$52.05

South Australia ≈ \$63.14

Tasmania ≈ \$115.49

Australian Capital Territory ≈ \$128.51

Northern Territory ≈ \$152.05 **[4 marks]**

d The survey covered only the capital cities, and did not include Australians living in regional and rural areas, which are likely to have higher fees.

Telephone polls are taken on landlines to ensure the correct distribution, but many people now, particularly younger people, do not have landlines.

The person answering could be a child.

There will also be recall and non-response bias.

While the national figures for those in the capital cities are likely to be accurate, the much lower accuracies for smaller state and territories means that individual state and territory results and comparisons are not very accurate. **[3 marks: 1 for each point]**

[14 marks altogether]

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